

TANVIR HOSSAIN

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SUMMARY

Electrical Engineering Ph.D. candidate with hands-on digital design and simulation-based functional verification experience across the RTL-to-GDSII flow: RTL design in Verilog/VHDL, Verilog testbench development and functional verification (ModelSim), and synthesis and back-end with Cadence (Genus, Encounter). Research program centers on rigorous, specification-driven evaluation of integrated circuits at both block and system level, with strong scripting (Tcl, Bash, Python), C, and independent debug of complex digital designs. Seeking a Design Verification Engineer role to apply RTL verification, scripting, and debug skills to SoC and IP subsystem development.

TECHNICAL SKILLS

Design & Verification:	RTL design and simulation-based functional verification; Verilog testbenches; directed and self-checking test development; functional simulation and debug.
Hardware Description Languages:	Verilog, VHDL.
EDA / DV Tools:	ModelSim (functional simulation); Cadence Genus, Encounter, Virtuoso; PSpice, HSPICE.
FPGA & Prototyping:	Xilinx Vivado, ISE, SDK; Intel Quartus. Boards: Basys 3 Artix-7, Nexys A7-100T, Altera DE2-115, HaHa (Intel MAX10).
Programming:	C, Python, MATLAB, Assembly (MIPS); LLVM toolchain.
Scripting:	Tcl, Bash, Git.
Architecture & Modeling:	gem5, McPAT, Intel VTune; computer architecture, memory (SRAM/RRAM), and bus/datapath fundamentals.
Lab & Measurement:	Tektronix DPO70404C oscilloscope, PicoScope, ChipWhisperer; embedded targets (Raspberry Pi, Arduino, STM32, ESP32).

RELEVANT EXPERIENCE

The University of Kansas <i>Graduate Research Assistant, Digital Hardware Design & Security</i>	Jan 2022 – Present Lawrence, KS
- Design and verify digital IP at block and system level using Verilog/VHDL, driving designs through RTL, simulation-based functional verification (ModelSim), and Cadence synthesis (Genus, Encounter). - Develop directed and self-checking testbenches to exercise designs against their specifications, analyze simulation failures, and independently debug functional issues down to root cause. - Build and prototype hardware on FPGA platforms (Artix-7, Intel MAX10, DE2-115), automating design, simulation, and analysis flows with Tcl, Bash, and Python. - Published verification and security evaluation results in IACR TCHES, ACM TODAES, and the Journal of Hardware and Systems Security.	
Keysight Technologies <i>Research Intern, Applied ML & Hardware Security</i>	May 2026 – Aug 2026 Santa Rosa, CA
- Built model-guided optimization and adaptive-learning workflows for fault-injection parameter tuning, including experiment design, telemetry instrumentation, scalable hyperparameter search, and reliability metrics. - Delivered a prototype demonstration targeting improved vulnerability-detection coverage over baseline flows.	
The University of Kansas <i>Graduate Teaching Assistant, EECS</i>	Aug 2024 – Dec 2024 Lawrence, KS
Led labs and graded for digital logic and computer architecture; guided project-based design using MARS (MIPS) and gem5 (x86) simulators.	

EDUCATION

Ph.D. in Electrical Engineering , The University of Kansas <i>Advisor: Dr. Tamzidul Hoque</i>	Expected 2026
M.S. in Electrical Engineering , The University of Kansas	May 2025
B.Sc. in Electrical and Electronic Engineering , Ahsanullah University of Science and Technology, Bangladesh	2016 – 2021

SELECTED PROJECTS

- **RTL Evaluation of Design-for-Security Scan & PUF Primitives:** Modeled and verified Ring-Oscillator PUFs and Dynamically Obfuscated Scan Chains (DOSC) at RTL, then injected and simulated targeted modifications to characterize behavior at block and system level under directed stimulus — exercising scan-chain and authentication logic against specification.
- **Multi-Valued Logic Circuits for Hardware:** Designed and simulated multi-valued logic circuits, analyzing functional behavior and trade-offs; published at IEEE DCAS 2023.
- **GNRFET-Based 6T/7T SRAM Cells:** Performed transistor-level design and SPICE simulation of SRAM bit-cells, characterizing read/write behavior and stability across variants.
- **FPGA Hardware Teaching Platform:** Built Verilog/FPGA and ESP32 design modules and demos used in NSF-funded curricula, validating each on Artix-7 and Intel MAX10 boards.

SELECTED PUBLICATIONS

Selected from 17 peer-reviewed publications. Name in **bold**. Full list: [Google Scholar](#).

1. **Tanvir Hossain**, S. M. Mojahidul Ahsan, and Tamzidul Hoque, “Potential and Pitfalls of Multi-valued Logic Circuits for Hardware Security,” *2023 IEEE 16th Dallas Circuits and Systems Conference (DCAS)*, 2023. DOI: [10.1109/DCAS57389.2023.10130261](#).
2. S. M. Mojahidul Ahsan, **Tanvir Hossain**, Md Sakib Hasan, and Tamzidul Hoque, “Resistive RAM-based PUF: Challenges and Opportunities,” *2023 IEEE 16th Dallas Circuits and Systems Conference (DCAS)*, 2023. DOI: [10.1109/DCAS57389.2023.10130179](#).
3. S. M. Mojahidul Ahsan, M. S. Shahriar, M. Chowdhury, **Tanvir Hossain**, M. S. Hasan, and Tamzidul Hoque, “Accurate, Yet Scalable: A SPICE-based Design and Optimization Framework for eNVM-based Analog In-memory Computing,” *43rd IEEE/ACM ICCAD*, 2024. DOI: [10.1145/3676536.3676827](#).
4. **Tanvir Hossain**, Md Shyeem Rahman, Md Mushfiqur Rahman, and A. R. A. Dibbo, “Performance Analysis of GNRFET-based 6T and 7T SRAMs,” *2020 23rd ICCIT*, 2020. DOI: [10.1109/ICCIT51783.2020.9392713](#). (**Best AUST Paper Award**)
5. **Tanvir Hossain**, S. M. M. Ahsan, P. Chakraborty, S. Bhunia, and Tamzidul Hoque, “Attacking the Protections of Hardware using Trojan-Assisted Meta Attacks,” *ACM Transactions on Design Automation of Electronic Systems (TODAES)*, 2026. DOI: [10.1145/3804452](#).
6. **Tanvir Hossain**, Matthew Showers, Mahmudul Hasan, and Tamzidul Hoque, “On the Security of Lightweight Homomorphic Obfuscation for Protecting Against Hardware Trojans,” *IACR Transactions on Cryptographic Hardware and Embedded Systems (TCHES)*, vol. 2026, no. 2, 2026. DOI: [10.46586/tches.v2026.i2.736-763](#).

SELECTED HONORS & AWARDS

MAY 2023	Champion, Microelectronics Security Challenge (IP security) — IEEE HOST 2023 Symposium
2022–2026	NSF Travel Grants — IEEE HOST Symposium (2022, 2023, 2026)
APRIL 2023	David D. and Mildred H. ROBB Award — EECS Department, University of Kansas
JAN 2024	First Place, Poster Presentation — I2S Student Research Symposium
MARCH 2021	Best AUST Paper Award — ICCIT 2020